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ORIGINAL PAPER

Automatic identification of mining-induced subsidence using deep convolutional networks based on time-series InSAR data: a case study of Huodong mining area in Shanxi Province, China

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Abstract
Underground mining has the potential to trigger wide-area surface subsidence, and it is crucial to identify and investigate mining-induced subsidence to reduce potential damages. In recent years, the development of remote sensing and deep learning has brought a novel direction to the automatic identification of mining-induced subsidence. In this paper, we present a comprehensive solution for the automatic identification of mining-induced subsidence using deep convolutional networks based on time-series interferometric synthetic aperture radar (InSAR) data. First, we obtain a surface deformation rate map of the Huodong mining area based on SBAS-InSAR technology. Second, we automatically identify mining-induced subsidence areas from the InSAR deformation map based on a convolutional neural network (CNN) deep learning model. Finally, we summarize the potential risk of surface subsidence to roads in the Huodong mining area. The InSAR results show the detailed characteristics of surface deformation with a maximum subsidence rate of 112.1 mm/year, where the surface deformation is distinctly different between the urban, forest, and mining areas. The deep learning results show 202 identified mining-induced subsidence areas, including the subsidence areas located in the forest which were difficult to distinguish in the InSAR monitored deformation map. This paper demonstrates the potential and capability of deep learning models for more automatic and accurate identification of mining-induced subsidence and provides a comprehensive solution for improving the efficiency of interpreting surface deformation monitoring information.

Keywords Surface subsidence · Underground mining · Deep learning · InSAR monitoring · DeepLabv3 +

Introduction

Underground mining activities worldwide have triggered wide-area surface deformation. Mining-induced surface deformation commonly leads to a series of geohazards, such as landslides and ground subsidence, in addition to creating difficulties in engineering construction and urban development (Whittaker and Reddish 1989; Altun et al. 2010). Consequently, it is essential to develop high spatial and temporal resolution surface deformation monitoring technology and accurate and efficient deformation information analysis approaches.

Recent advances have demonstrated the significance of remote sensing technology in surface deformation monitoring, especially using synthetic aperture radar interferometry (InSAR) technology (Osmanoglu et al. 2016; Yang et al. 2020a, b). InSAR monitoring overcomes the disadvantages of traditional monitoring technologies, such as high costs, observations being based on discrete points, and susceptibility to weather factors. Specifically, InSAR provides the ability to monitor surface deformation using Sentinel-1 images with a width of 250 km and a spatial resolution of 5 m × 20 m (Geudtner, Torres et al. 2014). Currently, InSAR technology has been applied in urban construction (Dong et al. 2014), disaster prevention (Zhao et al. 2012), mine management, and especially in large-scale surface deformation monitoring (Liu et al. 2013).

The application of small baseline set (SBAS) technology integrated with InSAR can further improve the time-series

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monitoring capabilities of InSAR (Bateson et al. 2015). In recent years, the development of SBAS-InSAR technology has become one of the most advanced methods for continuous ground deformation monitoring. In an earlier investigation, Chaussard et al. (2013) used SBAS-InSAR technology to monitor time-series surface deformation in several cities in western Indonesia. The following year, they monitored ground deformation in central Mexico based on ALOS data from 2007 to 2011 and identified groundwater extraction as the main cause of deformation (Chaussard et al. 2014). In addition, Grzovic and Ghulam (2015) employed SBAS technology to assess ground subsidence caused by underground coal mining in Springfield, Illinois, and the study demonstrated that another advantage of remote sensing is the detection of regions of unknown deformation.

InSAR technology has been able to obtain large-scale and high-precision time-series surface deformation data; however, the deformation obtained by single InSAR technology is commonly triggered by various factors, such as underground mining, vegetation growth, and other human activities. Underground mining-induced subsidence has unique spatial, geometric, and deformation characteristics (Fan et al. 2021), and accurate monitoring of mining subsidence is a scientific basis for the government to monitor illegal mining activities. Actually, it has been difficult to learn the complex features of mining-induced subsidence from InSAR data to achieve accurate identification based on the traditional analysis methods (Ng et al. 2012; Dong et al. 2015; Li et al. 2015). For example, when investigating the characteristics of high-intensity mining subsidence in the Shendong coalfield, Ma et al. (2016) discovered that it was difficult to distinguish surface deformation associated with past mining activities and atmospheric influences based on time-series analysis.

Currently, the rapid development of deep learning has brought a novel solution to the mining and analysis of InSAR data (Ma and Mei 2021; Ma et al. 2022). Recent studies on deep learning applied to InSAR data analysis have mainly focused on using convolutional neural networks (CNNs) for phase unwrapping of SAR interferograms (Rotter and Muron 2021). For example, Wu et al. (2022) proposed a deep learning strategy to obtain time-series ground deformation from large-scale noise-contaminated interferograms; however, the proposed method was unable to classify the deformation zones induced by different factors. In applying deep learning to analyze the deformation features induced by different factors (Wang et al. 2022b), Anantrasirichai et al. made a new attempt (Anantrasirichai et al. 2019; Anantrasirichai et al. 2021); they developed an AlexNet model with synthetic interferograms as training data to detect ground deformation related to volcanic activity from InSAR data; however, the synthetic interferogram simplifies the real signal and causes unsatisfactory prediction accuracy in practical applications.

Recent studies have indicated the potential of deep learning to learn complex characteristics of surface deformation from InSAR data, and real training data is a huge challenge using CNN models to achieve automatic identification. In our previous research (Liu et al. 2021), we investigated the surface deformation of the Yangquan mining area in Shanxi Province using InSAR monitoring and obtained large-scale time-series InSAR data. There are a huge number of mining areas distributed in Shanxi Province, and underground mining is the main cause of ground deformation. The acquired InSAR data of the Yangquan mining area from our previous research work can offer training data of deep learning models to more efficiently learn features of mining-induced subsidence for application of automatic identification of mining-induced subsidence in other mining areas, thus contributing to the prevention of mining disasters in Shanxi Province and serving as a scientific basis for the government to monitor illegal mining activities.

Motivated by this, this paper presents a comprehensive solution to the automatic identification of mining-induced subsidence using deep convolutional networks based on time-series InSAR data in the Huodong mining area, another typical mining area in Shanxi Province (Shi et al. 2021). The objective is to demonstrate the potential and capability of deep learning models for more automatic and accurate identification of mining-induced subsidence from time-series InSAR data. Specifically, we first obtained a surface deformation rate map and investigated the detailed characteristics of surface deformation in the Huodong mining zone based on SBAS-InSAR technology; second, we automatically identified mining-induced subsidence areas based on a CNN deep learning model from InSAR monitoring results; third, we summarized the potential risk of surface subsidence to roads in the Huodong mining area.

The remainder of the paper is organized as follows. The "Materials and methods" section introduces the study area and the automatic identification methods in detail. The "Results" section describes the results of InSAR monitoring and deep learning models. The "Discussion" section discusses this paper. The "Conclusion" section concludes this paper.

Materials and methods

Study area: Huodong mining area in Shanxi Province, China

The Huodong mining area is located in Shanxi Province, China, between Linfen city and Changzhi city; see Fig. 1. The mine area is approximately 111.20 km long from north to south and 50.33 km wide, bounded between 36°21′01″N ~ 37°00′46″N from north to south and 111°58′45″E ~ 112°18′26″E from east to west, with an area of 1460 km^2. The area is mainly

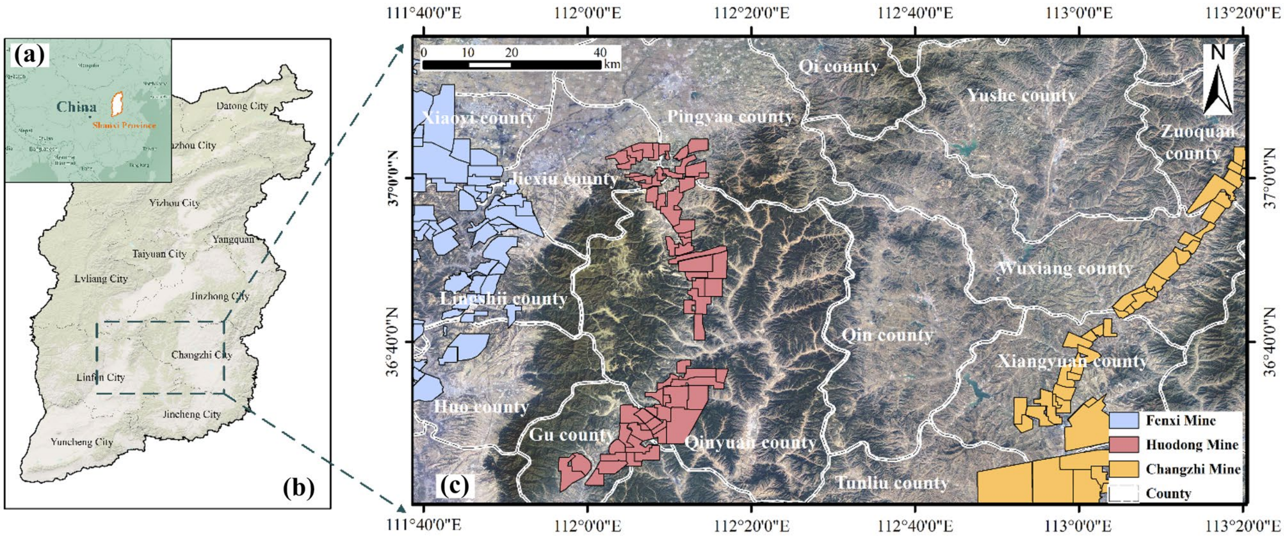

Fig. 1 The location of the study area in Shanxi Province, China

accessible by road, with National Highway 309 passing through the southern end of the mine, the South Tongpu Railway running 68 km west along National Highway 309 to Ganting Town, Huodong County, and the Dayun Expressway to the north and south of Ganting Town. The SAR image data selected for this application include the Huodong mining area, as well as the eastern side of the Fenxi mining area and the northern side of the Changzhi mining area.

The Huodong mining area has abundant mineral resources, including coal, bauxite, dolomite, iron, limestone refractory clay, and many other minerals. At present, the minerals that have been developed and utilized are coal, iron, limestone, building stone, river sand, groundwater, and brick and tile clay of seven kinds. A total of 65 coal mines have been built in the mining area. As shown in Fig. 2, due to the relatively backward mining technologies employed and a general lack

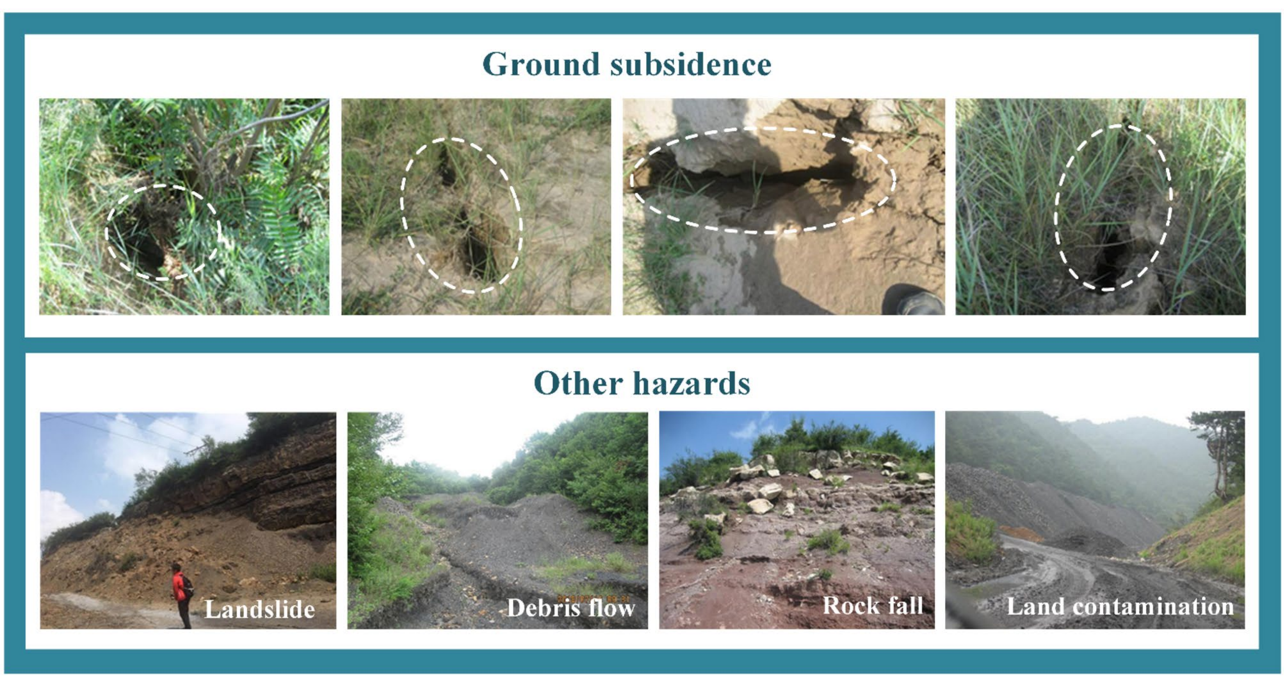

Fig. 2 Geological hazards in the study area

of environmental awareness, ground subsidence, debris flow, rock fall, landslides, and other disasters have occurred from time to time. Mining exploitation will remain the focus of national economic development in the Huodong mining area for decades to come. Therefore, it is essential to investigate the subsidence characteristics in the Huodong mining area for early warning of potential damages.

Methods

This paper presents a comprehensive solution to the automatic identification of mining-induced subsidence using deep convolutional networks based on time-series InSAR data in the Huodong mining area. First, we acquired the surface deformation rate map of the Huodong mining area based on SBAS-InSAR technology; second, we preprocessed the InSAR monitored data of the Yangquan mining area obtained from our previous work as training data of deep leaning model; third, we established a CNN deep learning model for automatic identification of mining-induced subsidence areas; finally, we applied the established model to identify mining-induced subsidence areas form InSAR monitored data of the Huodong mining area. The main process is shown in Fig. 3.

Step 1: acquiring surface deformation data based on the SBAS-InSAR technology

InSAR is a common monitoring technology that uses the phase information contained in SAR radar images as the basis for a comprehensive calculation to obtain spatial and temporal information on the surface. The core idea of InSAR is to interfere with two images by imaging the same surface target twice and then calculating the phase of each observation, solving for the interferometric phase difference, and inverting the terrain and deformation data.

In this paper, SBAS-InSAR technology is used to acquire surface deformation information in the study area. SBAS-InSAR technology can efficiently reduce the effects caused by the lack of spatial and temporal coherence (Yu et al. 2019). Compared with other InSAR methods, such as D-InSAR (Xia et al. 2018) and PS-InSAR (Hooper et al. 2012), the SBAS processing technique generates more interferometric image pairs within the relevant threshold range for the same number of SAR images. For image processing, first, spatial and temporal baselines are set; image pairs that meet the constraints are selected within the dataset; then, differential interferometric processing is performed; and finally, the pixel points with

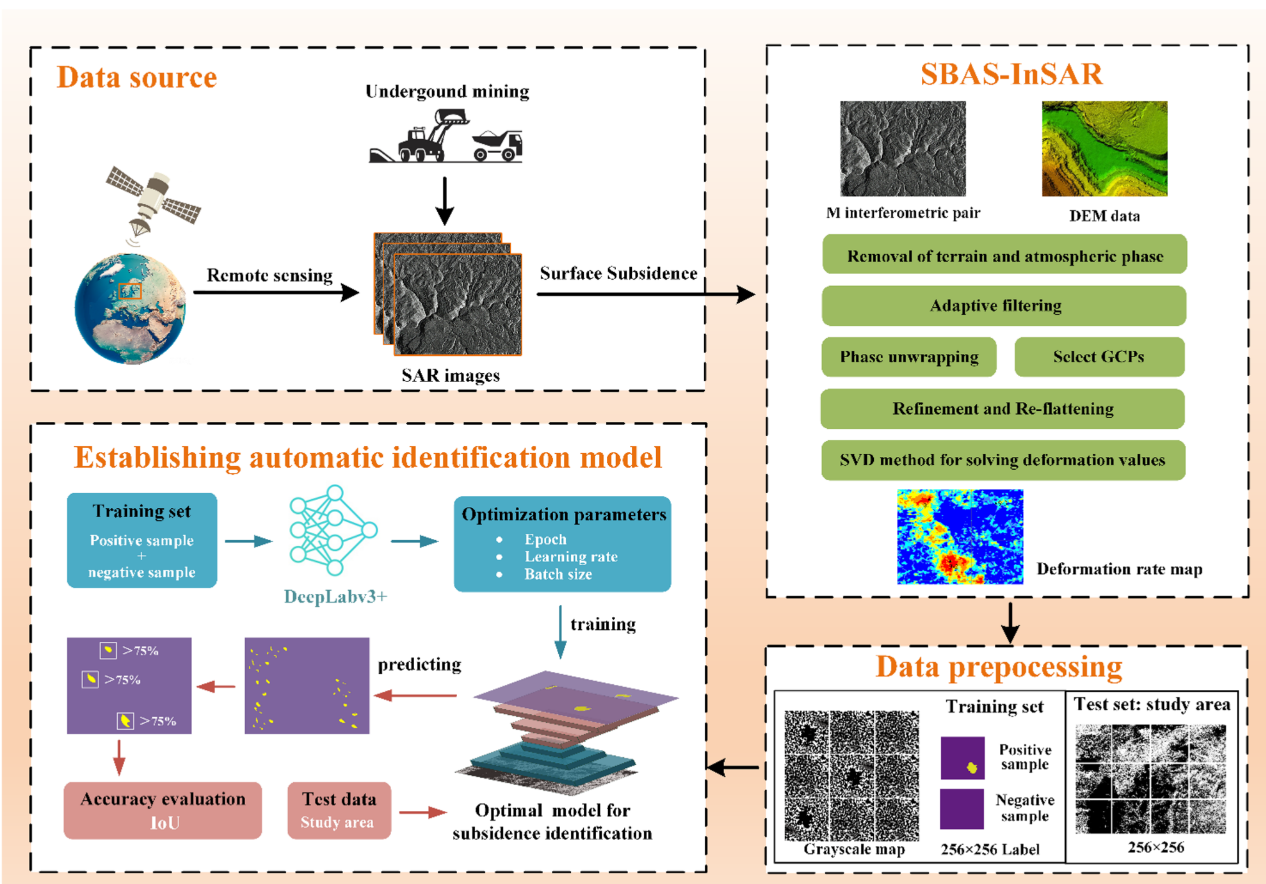

Fig. 3 Workflow of this paper

high coherence in the interferogram are used as the basis for inversion to obtain time-series deformation information. The SBAS-InSAR processing process for the Huodong mining area in this paper is specified as follows.

- Setting the time and space baselines. $N + 1$ SAR images are acquired for all times in the same area, from which M pairs of interferograms with baselines below the threshold are filtered out. M pairs satisfy the conditions shown in Eq. (1).

$$\frac{N}{2} \leq M \leq \frac{(N-1)N}{2} \tag{1}$$

- Aligning with external digital elevation model (DEM) data and precision orbital data to generate M multitemporal differential interferograms by removing topographic phases and atmospheric delay phases from M sets of interferometric image pairs. In i interferometric phase diagram, the interferometric phase representation of the pixel points with azimuth and distance (r, x) is shown in Eq. (2), where $\left(t_A, r, x\right)$ and $d\left(t_B, r, x\right)$ are the cumulative deformation of t_A and t_B ($t_A < t_B$), respectively; $\varphi\left(t_A, r, x\right)$ and $\varphi\left(t_B, r, x\right)$ are the cumulative deformation phases of t_A and t_B, respectively; and λ is the signal wavelength.

$$\delta\varphi_i(r,x) = \varphi\left(t_B, r, x\right) - \varphi\left(t_A, r, x\right) = -\frac{4\pi}{\lambda}\left[d\left(t_B, r, x\right) - d\left(t_A, r, x\right)\right] \tag{2}$$

- Transforming the interferometric phase expression into the form of the average phase velocity multiplied by the time baseline, as shown in Eq. (3), where $v_i(r, x)$ denotes the average phase velocity of i interferogram.

$$v_i(r,x)\left(t_B - t_A\right) = \varphi\left(t_B, r, x\right) - \varphi\left(t_A, r, x\right) \tag{3}$$

- Establishing the observation equation between the phase velocity of each cycle and the M differential interference phase by combining Step 2 and Step 3, as shown in Eq. (4), where B is the $M \times N$ matrix consisting of 0, 1, -1; v is the vector representation of the phase velocity at each cycle; and $\delta\varphi$ is the vector representation of the M differential interference phase.

$$Bv = \delta\varphi \tag{4}$$

- Employing the Singular Value Decomposition (SVD) method to further obtain the velocity vector and finally calculate the time-series surface deformation value by integration into the time period (Dong et al. 2014).
- Adding the projection information, such as latitude and longitude, to obtain the final and complete surface deformation rate map in the study area.

Step 2: preprocessing InSAR monitored data for deep learning model

In this paper, an automatic identification model is trained based on deep learning semantic segmentation approach to accurately and rapidly identify subsidence areas induced by underground mining from the surface deformation rate map monitored by InSAR. Before applying the deep learning model to identify mining-induced subsidence in the Huodong mining area, a large amount of training data for the deep learning model needs to be acquired.

Yangquan mining area is located in the southwestern of Yangquan City, Shanxi Province, covering an area of about 2102.47 km^2, which is the largest coal-producing area in Yangquan City. In our previous research work, we monitored the time-series surface deformation based on SBAS-InSAR technology and obtained a surface deformation rate map of the Yangquan mining area by Step 1. This paper employed obtained surface deformation rate map of the Yangquan mine (Liu et al. 2021) as training data to train the deep learning model and make predictions in the Huodong mining area.

Before training the deep learning model for the automatic identification of mining-induced subsidence areas, the obtained surface deformation rate map of the Yangquan mining area needs to be preprocessed to produce balanced positive and negative samples. The specific processing steps for training data in this paper are shown in Fig. 4, and finally, a total of 10,000 training images were produced.

1. Data format conversion. The deformation values contained in the original InSAR surface deformation rate map (RGB) are converted to gray values (0–255) in a standard image to adapt the input of the deep learning model.
2. Image segmentation. We standardly segment the processed grayscale map into image patches with the size of 256×256 based on 50% coverage.
3. Positive and negative sample production. Image patches that contain mining sinkholes are positive samples, and samples that do not contain mining sinkholes are negative samples. The positive and negative samples are manually labeled based on mining statistics and ground hazard survey information of the Yangquan mining area. As shown in Fig. 4, the sample patches are binary images, which use purple pixel points (value 0) to represent non-mining-induced subsidence areas and yellow pixel points to represent mining-induced subsidence areas.
4. Data enhancement. It is important to balance the number of positive and negative samples in a deep learning model. In this paper, we expand the number of positive

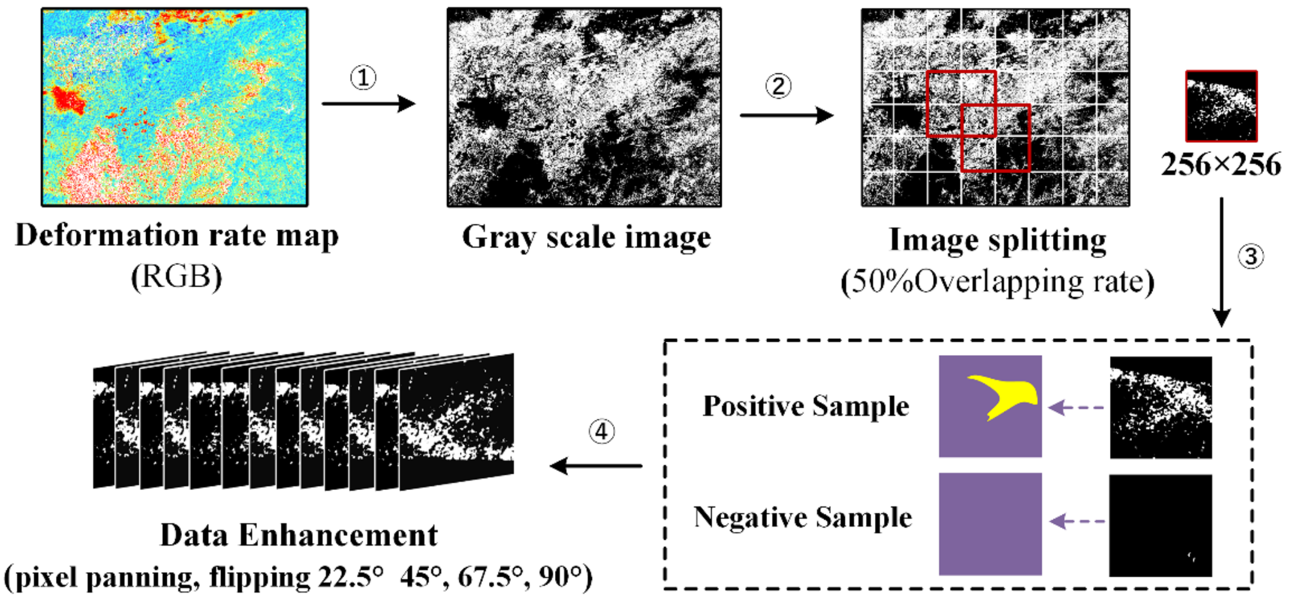

Fig. 4 Preprocessing process of InSAR deformation images

and negative samples by pixel panning, horizontal and vertical flips, rotations of 22.5°, 45°, 67.5°, and 90°, and different scale twists along the radially varying x and y coordinate axes. We finally obtained a total of 10,000 training samples.

Step 3: establishing deep learning model for automatic identification of mining-induced subsidence areas

This paper developed a CNN deep learning model for the automatic identification of subsidence areas induced by underground mining. Convolutional neural networks (CNNs) are a common type of deep learning models, typically consisting of a convolutional layer, a pooling layer, and a fully connected layer. Compared with other deep learning classification models, CNNs use a strategy of shared convolutional kernels, which allows for the efficient processing of high-dimensional data using fewer training parameters, provides strong generalization ability for the models produced, uses pooling operations to reduce the spatial dimensionality of the network, and provides more relaxed requirements for the input data.

DeepLabv3 + is one of the most successful semantic segmentation models based on CNNs (Chen et al. 2018). In the DeepLabv3 + model, the encoder part mainly consists of a backbone network and an atrous spatial pyramid pooling (ASPP) model, which mainly serves to sample the output feature map obtained, while the decoder is mainly used to recover the boundaries of the monitored image. The network uses cavity convolution to achieve feature extraction from the encoder at arbitrary resolutions, thus enabling the network to better balance monitoring speed and accuracy. In this paper, we conducted the automatic identification of mining-induced areas based on the DeepLabv3 + deep learning model. The main structure of the DeepLabv3 + model is shown in Fig. 5a.

When constructing the encoder of the DeepLabv3 + model for automatically identifying subsidence areas induced by underground mining, the ResNet network is employed as the backbone network used to extract features. The ResNet network (Wang et al. 2022a) inserts the residual learning module in the convolutional layer, which solves the problems of gradient disappearance, explosion, and network degradation that occur in traditional convolutional neural networks as the number of network layers deepens, making it easier to train the CNN models for image classification. The ResNet network is constructed as in Eq. (5), where x and y are the input and output vectors and f represents the residual mapping to be trained; W_1 and W_2 are the weights of the first and second layers of the convolution, and σ is the ReLU activation function.

$$y = f\left(x, \left\{W_i\right\}\right) + x$$

$$f\left(x, \left\{W_i\right\}\right) = W_2\sigma(W_1 x) \tag{5}$$

The structure of the ResNet-101 network constructed in this paper is shown in Fig. 5b. First, a convolutional layer with 7 × 7 × 64 inputs is set up; then, 33 (3 + 4 + 23 + 3) blocks are set up, where each block is divided into three layers, for a total of 99 layers; finally, fully connected layers are connected to output the classification results.

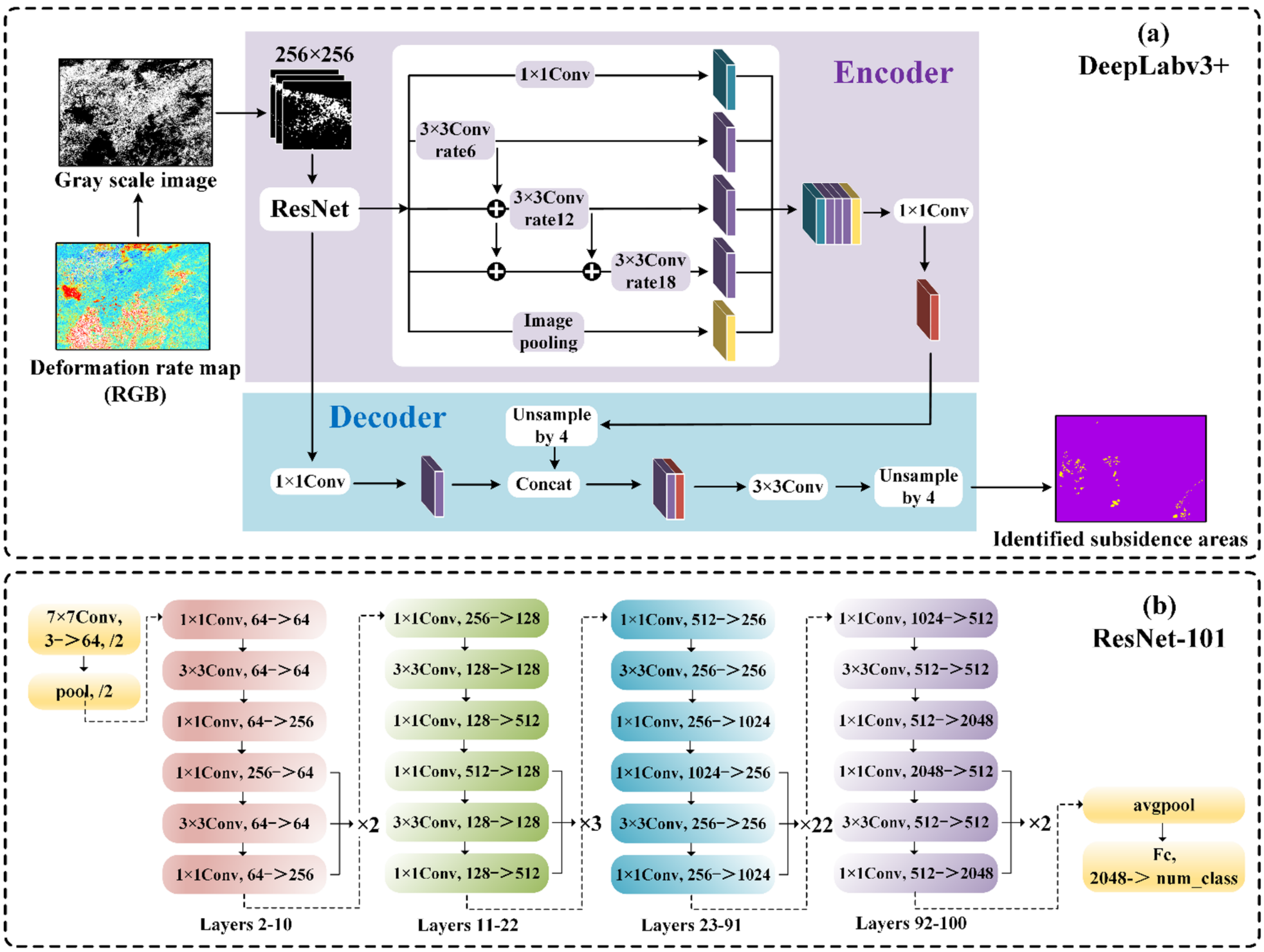

Fig. 5 DeepLabv3 + semantic segmentation model and ResNet-101 network

In the model training process, first, the preprocessed training samples of the Yangquan mining area are fed into the DeepLabv3 + network to learn the complex features of the mining-induced subsidence; then, the model parameters (epoch, batch size, and learning rate) are trained iteratively to obtain optimal parameters; finally, the deformation rate map of the study area is fed in for prediction. In the prediction process, the GDAL package (Bunting et al. 2014) in Python is used to directly input the complete InSAR deformation image without cropping, automatically intercepting blocks adapted to the model size by sliding recognition (with a 50% overlap rate), averaging the predicted probabilities for the overlapping parts, and resynthesizing the predictions into a complete result covering the study area.

Considering the disadvantages of incomplete edge recognition and noise in the DeepLabv3 + model recognition results, this paper uses strong deformation boundaries to identify InSAR surface deformation images and marks ground pixel points with confidence levels > 75% as underground mining-induced subsidence areas in the DeepLabv3 + prediction process.

Results

Investigated characteristics of surface deformation in the Huodong mining area based on the SBAS-InSAR technology

In this paper, SBAS-InSAR processing of remote sensing images of the study area was conducted using Envi-SARscape software. Images from RADARSAT-2, based on 18 views, were selected as the SAR imagery of the study area, spanning from 04/01/2017 to 10/07/2018, with a time baseline of 170 days and a spatial baseline of 350 m; a total of 48 interferometric pairs were combined.

After six steps of baseline estimation and connection map generation, interferogram generation, adaptive filtering and coherence generation, phase de-entanglement, track refinement, and redeplatforming and inversion and geocoding, we obtained the final complete surface deformation monitoring results of the study area. In particular, we performed two-track differential interference on SAR images based on orbital data and topographic data. In the differential interference process, multiview processing was used to reduce the difficulty of phase untwisting. The number of multiple views was set to 2×4, and the threshold of the coherence coefficient was set to 0.35. Then, time-series analysis was performed on the removed differential interference phases, and the residual phases were extracted using the spatial domain rate filtering method for atmospheric delay phases. Finally, differential interference was performed again to obtain the differential interferogram.

After processing by SBAS-InSAR, the time-series deformation values of the study area are obtained, and the average deformation rate is calculated and plotted as a surface deformation rate map, as shown in Fig. 6.

The InSAR monitoring results mainly cover the Huodong mining area as well as parts of the Fenxi mining area and parts of the Changzhi mining area. The results show that the larger areas of subsidence are mainly located in the northwestern part of the study area near the urban area of Jinzhong city and within the three mining claims, suggesting that surface subsidence is most likely due to underground mining. The maximum subsidence rate within the study area is 112.1 mm/year, with the maximum subsidence rate in the Fenxi mining area being 86.7 mm/year, the maximum subsidence rate in the Huodong mining area being 112.1 mm/year, and the maximum subsidence rate in the Changzhi mining area being 93.4 mm/year. The area with a subsidence rate exceeding 50 mm/year accounted for approximately 0.233% of the total study area.

In general, the subsidence rate within the mining area gradually decreases from the center to the edge, and the subsidence area is typically circular or elliptical in shape, with the absolute value of the gradient of the deformation value much larger than that of the nonsinkhole area and with the direction of the gradient points from the center of the subsidence area to the edge. In addition, apart from the mineral rights area, there are obvious subsidence areas located in the northwest region near the urban area of Jinzhong. Based on the located illegal mining areas using D-InSAR technology by (Xia et al. 2018), it can be tentatively concluded that the obvious subsidence areas outside the mineral rights area are caused by illegal mining.

In addition, based on the surface deformation data obtained from SBAS-InSAR technology and GIS visual interpretation, we investigated the deformation characteristics of the urban, forest, and mining areas in the study area, and the results showed that the surface subsidence of the three types of areas had distinct features, as shown in Fig. 7.

(a) The spatial distribution of the mining deformation zone is mostly funnel-shaped, with the maximum surface subsidence occurring mainly at the center of the mining area and the rate of subsidence decreasing from the center to the edge. The subsidence zone generally has a typical circular or elliptical shape, and the absolute value of the gradient of the deformation value is much greater than that of the nonsubmerged area, with the direction of the gradient pointing from the center to the edge of the subsidence zone.
(b) The spatial distribution of the urban deformation zone is irregular, with dramatic changes in deformation values. Generally, the deformation of built-up building plots is small and appears as light brown patches on the map, while the surface deformation of roads, green areas, and agricultural land appears to rise (dark green patches) or sink (light yellow patches) in an irregular manner due to human activities.
(c) The spatial distribution of the forest deformation zone unfolds in the direction of the forest extension, the deformation values vary strongly, and some of the pixel points that are too strong are rounded off during the interference process due to incoherence and are shown as null values and no pixel points in the map.

Identified mining-induced subsidence from InSAR monitored data of Huodong mining area based on the deep learning model

Identified results

In this paper, we developed an identification model of mining-induced subsidence areas based on the DeepLabv3+ model. The parameters of the training device and deep learning model are listed in Table 1.

A total of 202 subsidence areas induced by underground mining were identified by the DeepLabv3+ deep learning model, as shown in Fig. 8a. The identified underground mining-induced subsidence areas are mainly located in the Fenxi mining area, Huodong mining area, and Changzhi mining area. The spatial distribution of the identified underground mining-induced subsidence areas is mostly funnel-shaped, with large surface subsidence occurring mainly in the center of the mining area.

Identified accuracy

To evaluate the accuracy of the developed deep learning model in identifying underground mining-induced subsidence areas in the study area, we employed the intersection and union ratio (IoU) metric (Yang et al. 2020a, b), which is the ratio of the intersection and union sets of the actual and predicted values and is defined as shown in Eq. (6), where k

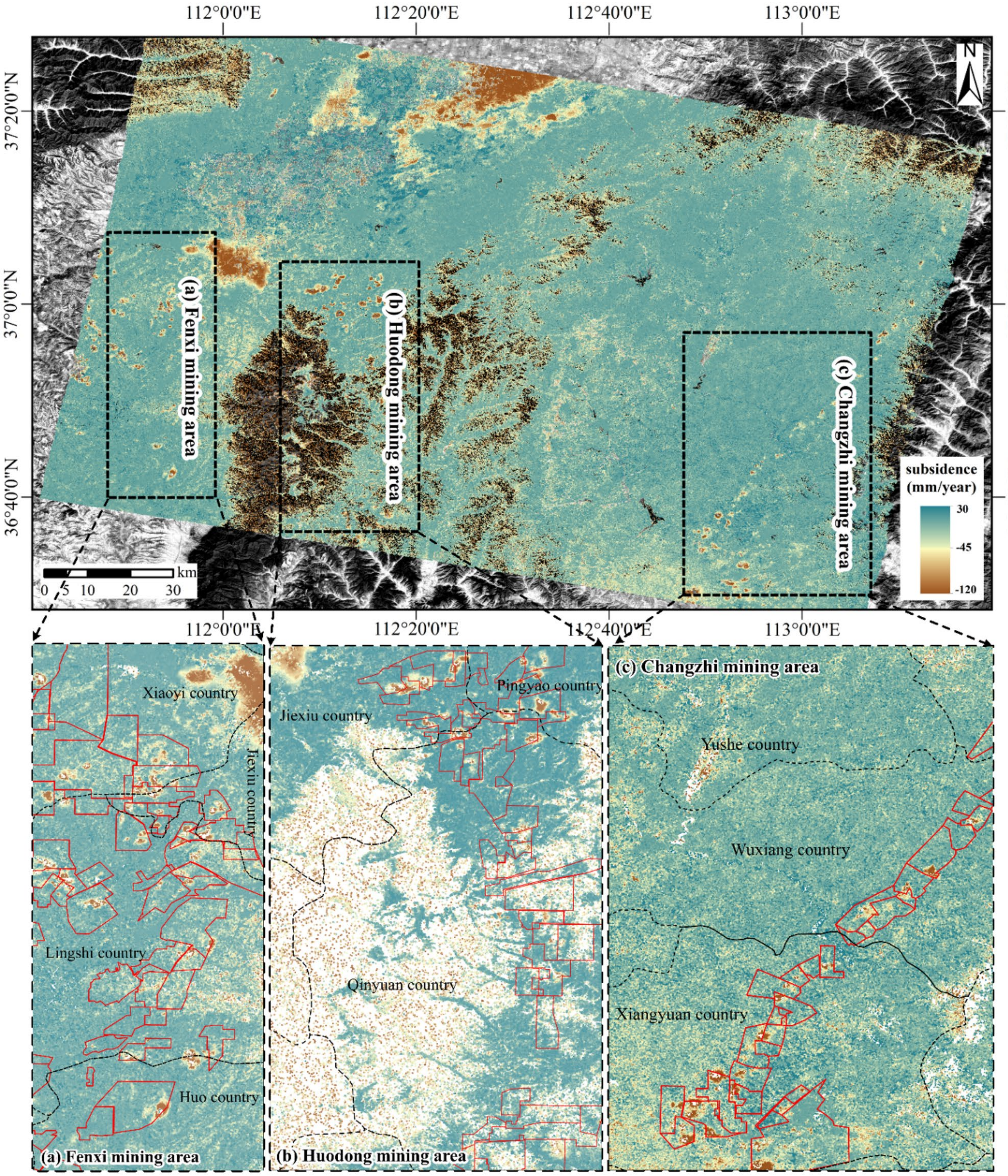

Fig. 6 Surface deformation rate map of the study area

is the number of categories specified in the training model (there is usually a background class; thus, there are a total of $k + 1$ categories in the model); C_{TP} indicates the number of correctly classified pixel points; C_{FP} indicates the number of false-positive pixel points; and C_{FN} indicates the number of false-negative pixel points.

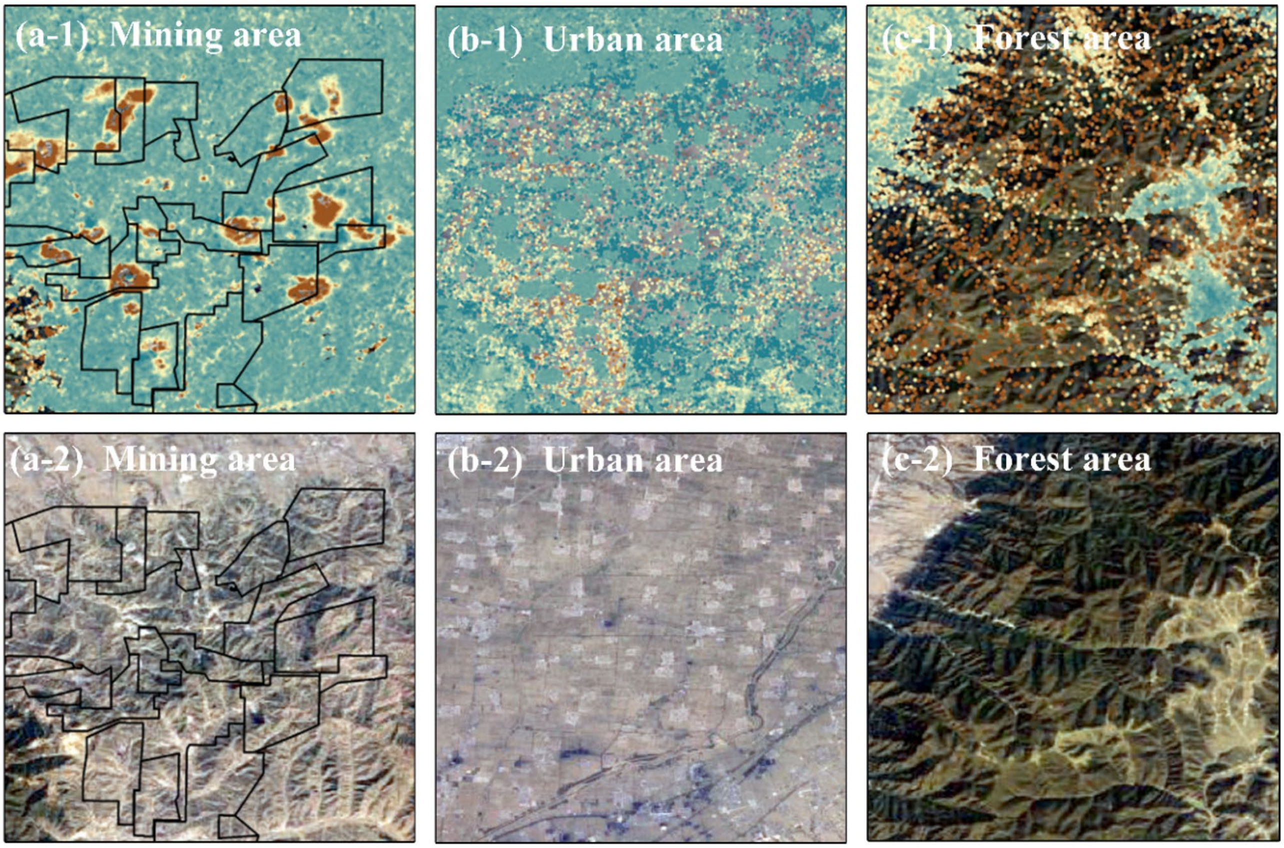

Fig. 7 Surface deformation characteristics of forest, urban, and mining areas

$$\alpha_{\mathrm{IoU}} = \frac{1}{k}\sum\nolimits_{k}\frac{C_{\mathrm{TP}}}{C_{\mathrm{TP}} + C_{\mathrm{FP}} + C_{\mathrm{FN}}} \tag{6}$$

After calculation, the IoU value of the DeepLabv3+ model for the underground mining-induced subsidence area in the Huodong mining area is 80.30%, which indicates that the DeepLabv3+automatic identification model for mining-induced subsidence areas developed in this paper is reasonably accurate.

To further refine the evaluation of the identification performance of the model, some identification results of the Changzhi mine area and Huodong mining area are extracted in this paper. Figure 8b shows the SBAS-InSAR monitoring results and deep learning model identification results for part of the Changzhi mine area; Fig. 8c shows the SBAS-InSAR monitoring results and deep learning model identification results for part of the Huodong mining area.

As shown in Fig. 8, to the east of the Changzhi mine is a forested area, which has varying degrees of deformation, as presented in InSAR monitoring results that will lead to confusion in the traditional visual interpretation process; this area is not flagged in the identification results based on the deep learning model. In addition, the area south of the Huodong mining area is forested, and seasonal variation in this area causes some of the image elements to be excluded from the InSAR monitoring results. Subsidence in this area is not easily observed via visual interpretation. The subsidence in the forest here is marked in the identification results of the developed deep learning model. All the above examples are further evidence of the effectiveness of the developed DeepLabv3+identification model.

Table 1 Parameters of the training device and deep learning model

Parameters of the device		Parameters of the deep learning model	
CPU-I5-9400F	6 cores, 6 threads	Epoch	80
GPU-NVIDIA-RTX2060	6G memory	Batch size	8
Maximum read speed	3500 MB/s	Learning rate	0.0001 in 1–25 epochs
Minimum write speed	2300 MB/s		0.00005 in 25–80 epochs

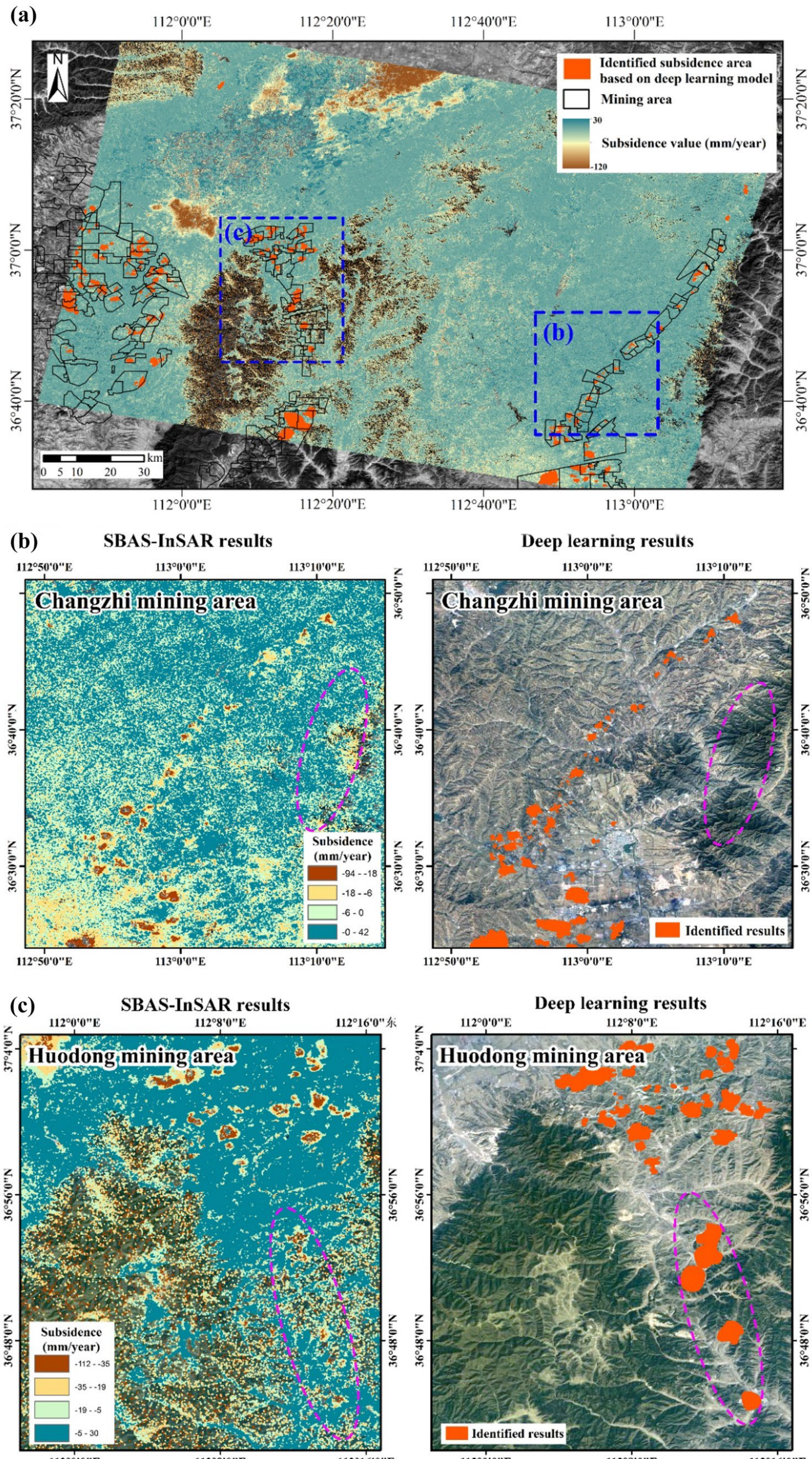

Fig. 8 The results of the identified mining-induced subsidence areas based on deep learning model

Analysis of potential risks on roads in the Huodong mining area

The surface subsidence induced by underground mining in the Huodong mining area has negatively impacted other industries in addition to threatening the mining industry. In recent years, with the development of the national economy and tourism, a road network system has been gradually built up with the counties as the center, connecting the surrounding counties and cities and radiating outward toward the towns and villages. National Highway 309 runs east–west through the southern part of the study area; Provincial Highway 323 runs diagonally through the mine area in the northern part of the study area, and Highway G55 runs north–south through the Changzhi mine area, covering a total of 370 km of graded roads in the study area. The study area is crisscrossed by large and small highways and railways, and with the continued development of the mining industry, surface subsidence caused by mining is bound to pose a risk to the roads along these routes.

In this paper, we investigated the potential risks caused by surface subsidence on roads in the study area based on the identification results of the developed deep learning model. In particular, we have annotated the dangerous road sections within the three mineral rights areas, and the results are shown in Fig. 9.

The subsidence area within the Fenxi mine area avoids all roads, and the surface subsidence induced by underground mining does not pose a threat to road construction there. The subsidence area within the Huodong mining area passes through a section of the S22 highway and two sections of the G241 highway, with a total length of approximately 5.19 km, of which the longest section is on the G241 highway, approximately 2.64 km in length. The subsidence area within the Changzhi mine area passes through a section of the G519 highway, two sections of the G55 highway, two sections of the G208 highway, and a section of the Y008 highway, with a total length of approximately 9.12 km. The longer two sections are the G208 highway, encompassing approximately 4.03 km. Railway construction and development will need to avoid these dangerous sections and take appropriate protection measures, as well.

Discussion

A short discussion on characteristics of surface subsidence induced by underground mining

Surface deformation results from multiple factors, such as underground mining, the cyclical growth of vegetation, or other human activities. Based on related studies of mining-induced subsidence (Bell et al. 2001; Ng et al. 2010; Ma et al. 2016; Salmi et al. 2017), it can be concluded that mining-induced subsidence areas have unique characteristics in terms of spatial, geometric, and deformational changes. As shown in Fig. 10, the specific characteristics are as follows.

Spatial characteristics: the maximum surface subsidence occurs mainly at the center of the quarry surface, with the rate of subsidence decreasing from the center to the edge, eventually forming a subsidence funnel at the surface of the quarry.

Geometric characteristics: the subsidence area is typically circular or oval in shape.

Deformation characteristics: the absolute value of the gradient in the subsidence zone is greater than that in the nonsinkhole zone, and the direction of the gradient runs from the center to the edge.

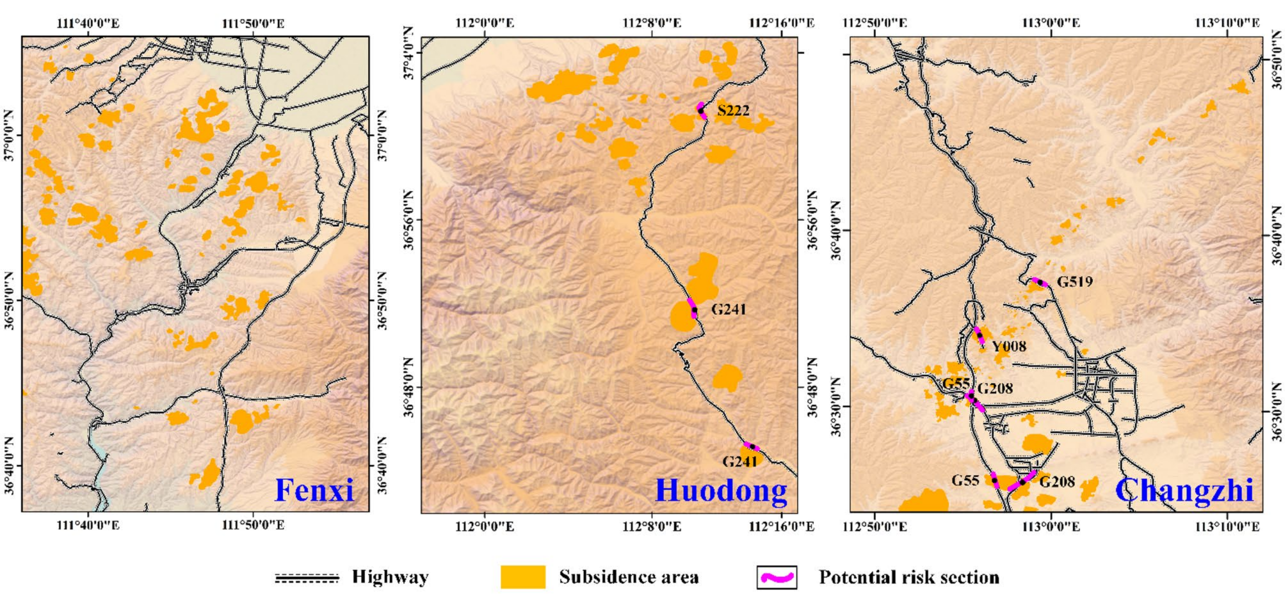

Fig. 9 The potential damage to sections of highways induced by surface subsidence

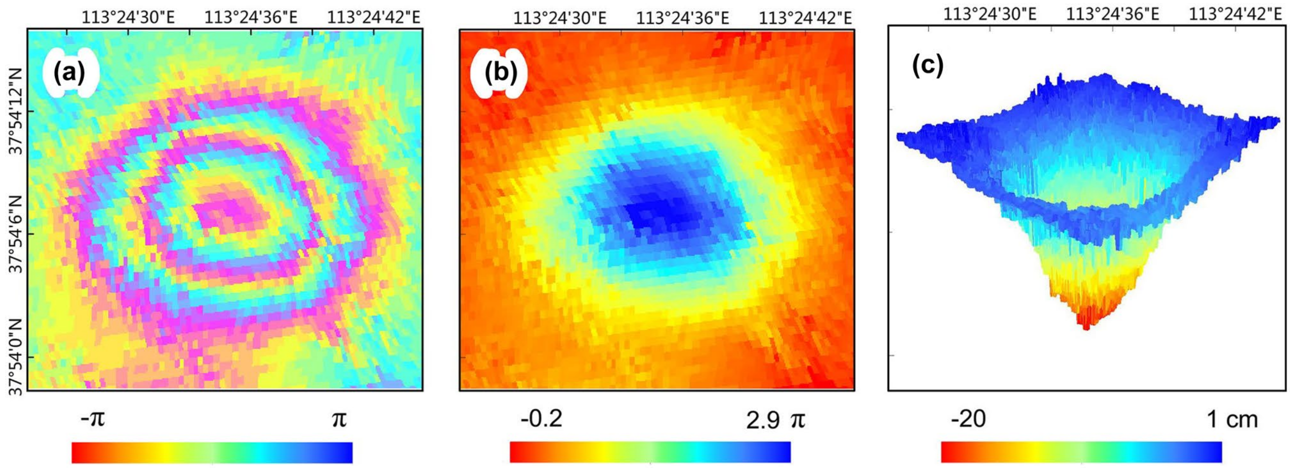

Fig. 10 Special features of subsidence areas induced by underground mining

In general, the traditional subsurface mining subsidence zone enclosure is mainly determined by using the shape and gradient of the subsidence zone to determine the specific causative factors of the subsidence zone, and the determination formula is shown in Eq. (7).

$$\mathrm{Cov}_m = a \times \mathrm{Cov}_{\mathrm{shape}} + b \times \mathrm{Cov}_{\mathrm{grad}} \tag{7}$$

where $\mathrm{Cov}_{\mathrm{shape}}$ is the shape factor; $\mathrm{Cov}_{\mathrm{grad}}$ is the gradient factor; a and b are the weights of the shape and gradient factors, respectively. A high correlation coefficient is calculated to identify a mining subsidence zone.

Currently, SAR satellite systems have the potential to monitor surface deformation in mining areas around the world. Sentinel satellites (Li and Roy 2017) provide an unprecedented amount of data for academics and government departments to use, providing a database to be used in real-time monitoring and providing warnings of potential disasters. At the same time, such an unprecedented amount of data is far beyond the scope of data that can be investigated by manual inspection, posing new challenges for disaster monitoring.

Obviously, traditional analysis methods have difficulty learning the complex features of subsidence induced by underground mining on the one hand; on the other hand, it is difficult to match the remote sensing big data. Therefore, some cutting-edge intelligent analysis methods are necessary for learning the subsidence characteristics induced by underground mining.

A comprehensive solution for mining-induced subsidence by combining InSAR and deep learning

The detailed characteristics of surface deformation in the study area can be obtained by SBAS-InSAR monitoring technology. However, the deformation obtained by single InSAR technology is commonly triggered by various factors, such as underground mining, vegetation growth, and other human activities. That is, single InSAR technology has limitations for identifying surface deformation induced by underground mining alone.

To obtain a comprehensive investigation of mining-induced surface deformation, this paper employed a cutting-edge deep learning model to learn the complex features of underground mining-induced subsidence from time-series InSAR data of the Yangquan mine area in Shanxi Province obtained from our previous research work (Liu et al. 2021) and then applied to identify mining-induced subsidence areas from the InSAR surface deformation map of the Huodong mining area and finally achieved accurate identification results. The presented results indicated that underground mining-induced subsidence in different mining areas has similar deformation characteristics, and deep learning has the ability to capture the complex features from the existing real time-series InSAR data to achieve automatic and accurate prediction in other similar scenarios.

More specifically, the identified results of the Huodong mining area showed that seasonal variations cause some of the pixel points in the forest areas in the surface deformation rate map to be missing, thus resulting in the uncertainty of whether deformation occurred in these areas based on single InSAR results, while the presented deep learning model automatically identifies mining-induced subsidence from these forest areas, which demonstrate that the deep learning model provides an accurate complement to the InSAR monitoring results. In the presence of the current proliferation of remote-sensing big data, deep learning has the potential to be one of the most efficient and intelligent methods for InSAR data analysis.

In summary, InSAR technology provides detailed surface deformation characteristics of the study area without determining the cause of deformation, while deep learning models are able to further identify mining-induced subsidence from InSAR monitored results automatically. This paper first investigates the detailed characteristics of surface subsidence in the Huodong mining area based on SBAS-InSAR monitoring technology, followed by a cutting-edge deep learning model to further identify mining-induced subsidence areas to complement the InSAR monitoring results. Finally, this paper summarizes the potential damages on the road in the study, providing a comprehensive investigation solution for mining-induced deformation by combining InSAR and deep learning.

For some districts where the mining industry is well developed, especially in Shanxi Province, illegal underground mining has caused a large number of coal mining accidents, and currently, most governments still use traditional microseismic or information network methods (Wang et al. 2022) to monitor illegal mining activities. The presented comprehensive solution in this paper serves a scientific basis for governments to monitor illegal mining activities more efficiently.

Outlook

In the future, much research work can be conducted. To begin, when using InSAR technology to investigate surface deformation for areas with dense vegetation or complex human activities, GIS remote sensing (Lu et al. 2019; Wang et al. 2022) deformation monitoring data or unmanned aerial vehicle (UAV) (Macrina et al. 2020) aerial photography data can be combined to enrich monitoring data sources and improve the monitoring accuracy. At the same time, deep learning models can be used to enable the monitoring equipment to identify surface activities autonomously and to correct the monitoring results to eventually eliminate human and natural disturbances from long-term remote sensing monitoring. However, when training the intelligent identification model to identify mining-induced subsidence areas, it will be necessary to expand the scope and number of training sets, perhaps using global or country-wide remote sensing data from international agencies, such as NOAA CLASS, ESA, and USGS Earth Explorer, to further improve the generality and accuracy of the method.

Conclusion

This paper presents a comprehensive solution for the automatic identification of the subsidence areas induced by underground mining using deep convolutional networks based on time-series InSAR data.

The monitoring results of the SBAS-InSAR technology in the study area show that (1) the high subsidence areas are mainly located in the three mining zones, with a maximum subsidence rate of 112.1 mm/year; and (2) the surface deformation of the three types of urban, forest, and mining areas has distinctly different features. The identified results of the deep learning model in the study area show that (1) there are 202 mining-induced subsidence areas, most of which are funnel-shaped, and that the maximum surface subsidence mainly occurs in the center of the mining zone; (2) the IoU identification accuracy of the deep learning model reached a high of 80.30%; and (3) the deep learning model can automatically identify subsidence areas located in forests which were difficult to distinguish in the InSAR monitoring results. The surface subsidence induced by underground mining has complex characteristics, and the utilization of deep learning can help improve the efficiency of automatic and accurate interpretation of surface deformation information.

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Data availability All data, models, or codes that support the findings of this study are available from the corresponding author upon reasonable request.

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